

Interpret inequalities



- 1
- a $n > 17$ b $n < 10$ c $w < \frac{2}{3}$ d $y \geq 12$
- e $6x \leq -42$ f $5 + 4x \geq 20$

- 2
- a p is less than 9. b f is greater than 14.
- c m is less than or equal to 23. d a is greater than or equal to 88.
-

Solve inequalities

3 -2

4 20

- 5
- a Is a solution. b Is a solution. c Is a solution. d Is not a solution.

- 6
- a 10 b Yes

- 7
- a 12 b Yes

- 8
- a 27 b Yes

- 9
- a Is not a solution. b Is a solution. c Is not a solution. d Is not a solution.

10 Yes, because we need to divide by a negative number.

11 She should have reversed the inequality symbol in step 3.

12

a $y > 6$

b $x \leq -6$



c $x > -7$

d $x \geq 4$

e $x < 16$

f $x > 5$

g $x \geq 8$

h $x > 6$

i $x \geq 13$

j $x > 5$

k $x \leq -7$

l $x \leq -3$

m $k \geq 4$

n $x > 2$

o $x \geq 21$

p $x \leq -32$

q $x \geq 63$

r $x < -6$

s $x \geq 6$

t $x > 0$

u $a < 66$

v $a > 0$

w $k \geq 9$

13

a $p < 5$

b $x \geq 4$

c $x \geq 1$

d $x \leq 10$

e $x > -\frac{3}{4}$

15

a $p \leq 4$

b $p = 4$

16

a $x \leq 1$

b

i Yes

ii No

iii Yes

iv Yes

17

a $x > 0$

b

i Yes

ii No

iii Yes

iv Yes

18

a $x \geq 3$

b

i No

ii Yes

iii Yes

iv No

Applications

19 $T < -15$

20 $x + 760 \geq 890$

21 Neville and Vincent

22 Tobias and Bob



23 $x \geq 87$

24 $p \leq \$27.90$

25

a $h \geq 46.31$

b 47 hours

26

a $w \geq 9.65$

b 10 windows

27

a $x + 520 \geq 740$

b No

28

a $f > 3m$

b Example answer: 14 female fish and 4 male fish.